

6. A) Differentiate between 'price (tax or subsidy) and quantity (regulation) approaches to addressing externalities.

Ans: Price (Tax or Subsidy) Approach:

1. Tax for Negative Externalities: When there is a negative externality (e.g., pollution), the government can impose a tax on the activity causing the externality. This tax raises the cost of engaging in the harmful activity, discouraging overproduction or consumption and moving the market toward a socially optimal outcome.
2. Subsidy for Positive Externalities: For activities that generate positive externalities (e.g., education or vaccination), the government can offer subsidies to lower the cost. This encourages more consumption or production of the beneficial activity, moving it closer to the socially optimal level.
3. Flexibility: The price-based approach allows market participants to decide how much they want to produce or consume based on the new price, giving them more flexibility and promoting efficiency through market forces.
4. Example: A carbon tax (price approach) incentivizes companies to reduce emissions in the way that is least costly to them.

Quantity (Regulation) Approach:

1. Regulations for Negative Externalities:
Governments can set strict limits or quotas on the amount of harmful activities (like emissions) allowed. This is a direct intervention that restricts the quantity of the externality-producing activity.
2. Mandatory Requirements for Positive Externalities:
In cases of positive externalities, the government may set minimum standards or requirements, such as mandatory vaccinations, to ensure that the activity is performed to a certain level.
3. Less Flexibility:
This approach is more rigid because it dictates a fixed quantity or standard, offering less room for market participants to adjust based on their individual circumstances.
4. Example: A cap-and-trade system (quantity approach) imposes a strict limit on emissions, but allows firms to trade permits, keeping the total emissions within the target limit.

In summary, the *price approach* adjusts incentives using taxes or subsidies, letting the market determine the quantity, while the *quantity approach* directly sets limits or mandates on the externality-producing activities.

B) Show that when firms have different marginal costs of pollution reduction, price mechanisms are more efficient means of accomplishing environmental goals unless quantity regulation is accompanied by the ability to meet regulatory targets by trading pollution permits across polluters.

Ans: To demonstrate why price mechanisms (e.g., taxes) are more efficient than quantity regulation when firms have different marginal costs of pollution reduction, let's break it down step by step.

Assumptions:

1. Firm A and Firm B have different marginal costs of reducing pollution.
2. The government wants to reduce total pollution to a certain level to reach an environmental goal.
3. We compare two approaches: (a) imposing a pollution tax (price mechanism) and (b) setting a strict pollution limit or quota (quantity regulation).

Marginal Cost of Pollution Reduction:

- Marginal cost of reduction (MCR) is the cost a firm faces to reduce an additional unit of pollution.
- If firms have different MCRs, it means one firm can reduce pollution more cheaply than the other.

1. Price Mechanism (Pollution Tax):

- Suppose the government imposes a tax T on each unit of pollution.
- Firms will individually decide how much pollution to reduce based on their marginal cost of reduction.
- Each firm will reduce pollution until their marginal cost of reduction equals the tax, i.e., until $MCR = T$.
- **Firm A (Low MCR)** will reduce pollution by more units because it's cheaper for it to do so.
- **Firm B (High MCR)** will reduce less because it's more expensive for it.
- This leads to an **efficient allocation** of pollution reduction across firms. The firms that can reduce pollution at a lower cost will do so more than the firms with higher costs. This minimizes the total cost of achieving the pollution reduction target, as firms equalize their marginal costs to the tax.

2. **Quantity Regulation (Cap without Trading)**:

- Suppose the government imposes a strict pollution limit or quota, requiring each firm to reduce pollution by the same amount.
- In this case, both **Firm A** and **Firm B** are required to reduce pollution by a fixed amount, regardless of their individual marginal costs.
- **Firm A**, with a lower MCR, ends up reducing pollution at a lower cost but is required to reduce just as much as **Firm B**, which faces a higher reduction cost.
- This results in an **inefficient outcome** because the total cost of pollution reduction is higher than necessary. If Firm B has a much higher MCR, it incurs greater costs to reduce the same amount of pollution, which could have been avoided if the reductions were distributed more efficiently.

3. **Quantity Regulation with Permit Trading (Cap-and-Trade)**:

- Now, suppose the government allows firms to **trade pollution permits** (cap-and-trade system).
- Firm A, with lower MCR, can reduce pollution more cheaply and sell excess pollution permits to Firm B, which faces higher costs of reduction.
- Firm B, instead of reducing pollution itself, buys permits from Firm A, allowing it to pollute more while staying within the overall pollution cap.
- **Trading leads to efficiency**: Total pollution is still reduced to the target level, but the firms that can reduce pollution more cheaply do so, minimizing the overall cost of achieving the environmental goal.

Conclusion:

- **Price Mechanism (Pollution Tax)** automatically leads to an efficient outcome, as firms equalize their marginal costs by reducing pollution until their MCR equals the tax.
- **Quantity Regulation without trading** forces all firms to reduce pollution by the same amount, which is inefficient when firms have different marginal costs.
- **Quantity Regulation with trading (Cap-and-Trade)** allows firms to trade permits, leading to an efficient allocation of pollution reduction similar to the price mechanism.

Thus, price mechanisms are more efficient unless quantity regulation is accompanied by trading.

C) Evaluate the market for pollution reduction with uncertain costs by considering two extreme cases of externalities: global warming and nuclear leakage.

Ans: When evaluating the market for pollution reduction under **uncertain costs**, the choice of approach (price mechanisms or quantity regulation) becomes critical. This analysis can be deepened by considering two extreme cases of externalities: **global warming** (a widespread, long-term externality) and **nuclear leakage** (a local, potentially catastrophic externality). Each of these presents different challenges in terms of uncertainty and the severity of outcomes, which affects the choice of policy.

1. **Global Warming: A Gradual, Global Externality**

Nature of Externality:

- Global warming is a widespread externality with long-term, cumulative effects that are not immediately observable.
- The impacts (e.g., sea-level rise, extreme weather) are distributed across countries and generations.
- There is uncertainty in the **marginal benefits of pollution reduction**, as the exact relationship between emissions and damage is hard to quantify in the short term.

Uncertain Costs of Pollution Reduction:

- Firms or countries face uncertain costs in reducing greenhouse gas (GHG) emissions due to varying energy prices, technological advancements, and future innovations in renewable energy.

Market Mechanism Evaluation:

- **Price Mechanism (Carbon Tax):** A carbon tax provides certainty in the cost of pollution but uncertainty in the quantity of emissions reduced. It gives firms flexibility to reduce emissions when it is most cost-effective for them.
 - Given the long-term and uncertain nature of the marginal benefits of reduction, a tax might be more efficient because it allows the market to allocate resources dynamically, as firms adapt to new technologies or price fluctuations over time.
 - **Example:** A carbon tax incentivizes firms to reduce emissions when it is cheapest to do so, minimizing economic disruptions while achieving gradual emissions reduction.
- **Quantity Regulation (Cap-and-Trade):** This provides certainty in the quantity of emissions but uncertainty in the cost of reducing emissions. If the cap is set too low or too high, it can lead to excessive costs or insufficient reduction.
 - For global warming, cap-and-trade systems are often used because they ensure a specific emissions target is met, even though it may lead to volatile permit prices.
 - **Example:** The European Union Emissions Trading System (EU ETS) sets an overall limit on emissions while allowing firms to trade permits, which helps reduce costs under uncertain market conditions.

Conclusion for Global Warming:

- **Price mechanisms (taxes)** might be more suitable given the uncertain costs and gradual, global nature of global warming. However, quantity regulation with trading can also be efficient if flexibility is introduced, such as adjusting caps based on market conditions.

2. **Nuclear Leakage: A Local, Catastrophic Externality**

Nature of Externality:

- Nuclear leakage is a highly localized, immediate, and potentially catastrophic externality with extreme consequences (e.g., radiation exposure, environmental contamination).
- The externality involves **low probability but extremely high-impact risks**, making it highly sensitive to even small amounts of pollution or accidents.

Uncertain Costs of Pollution Reduction:

- The costs of reducing risks from nuclear leakage (e.g., investing in safety technology, maintaining strict oversight) are uncertain, but the **marginal damage from accidents is catastrophic**. Any failure to adequately control leakage could lead to disasters with long-lasting effects on health, ecosystems, and property.

Market Mechanism Evaluation:

- **Price Mechanism (Pollution Tax or Risk Fee):** A tax on nuclear pollution would provide firms with flexibility to manage risks and invest in safety according to their cost structure. However, in the case of nuclear leakage, the uncertain but potentially catastrophic outcomes mean that even minor failures or misjudgments in risk reduction could lead to severe consequences.
 - A tax would not guarantee the strict controls needed to prevent such high-impact risks. Since the damage is not gradual but rather catastrophic, a tax could incentivize underinvestment in safety if firms underprice the risk of an accident.
- **Quantity Regulation (Strict Limits and Standards):** Due to the catastrophic potential of nuclear accidents, **quantity regulation** in the form of strict safety standards and operational limits is usually more appropriate.
 - Strict regulation ensures that firms adhere to specific safety protocols, such as maintaining certain levels of radiation containment, and avoids the risk of firms trading off safety for cost savings.
 - Given the **high marginal damage** of even small leakages, regulatory certainty in terms of maximum allowable emissions or operational safety is critical.
 - **Example:** Regulations such as those imposed by the International Atomic Energy Agency (IAEA) or local nuclear regulatory bodies ensure stringent controls over nuclear power plants and waste management, minimizing the probability of leakage.

Conclusion for Nuclear Leakage:

- **Quantity regulation** is the preferred approach because of the extreme risks involved. The costs of even a small failure are so high that flexibility in pollution reduction is not desirable. Strict, enforceable limits are necessary to minimize the probability of accidents.

Final Comparison:

- **Global Warming (Uncertain, Gradual):** Price mechanisms (e.g., carbon taxes) are more efficient because they allow flexibility and gradual adaptation to uncertainty in costs.

Cap-and-trade systems can also be effective if they allow permit trading to balance reduction efforts.

- ****Nuclear Leakage (Uncertain, Catastrophic)****: Quantity regulation is the best option due to the potentially catastrophic consequences of failure. Safety cannot be compromised for cost efficiency, so strict regulatory limits are necessary to manage risk.

In sum, the nature of the externality whether gradual and global or immediate and catastrophic—affects which policy is more efficient in the face of uncertainty.

7. (a) What are the informational asymmetries in the market models? Examine with examples.

Ans: Informational asymmetries in market models occur when one party in a transaction has more or better information than the other. This imbalance can lead to market inefficiencies and suboptimal outcomes. Here are some common types of informational asymmetries, along with examples:

1. Adverse Selection

Adverse selection happens before a transaction occurs when one party has information that the other does not. This often leads to the party with less information making suboptimal decisions.

Example:

Health Insurance: Potential policyholders know more about their own health status than the insurance company. Individuals with poor health are more likely to purchase insurance, leading to a pool of higher-risk individuals. This can result in higher premiums and may drive healthier individuals away from the market.

2. Moral Hazard

Moral hazard occurs after a transaction takes place when one party changes their behavior because they do not bear the full consequences of their actions.

Example:

Banking: After receiving a government bailout, a bank might engage in riskier financial activities, knowing that it might receive additional support if things go wrong. This behavior is different from what they would have exhibited without the safety net.

3. Signaling

Signaling is an action taken by the informed party to reveal some of their private information to the less informed party. This can help reduce informational asymmetry.

Example:

Education: A job candidate may obtain a degree from a prestigious university to signal their competence and intelligence to potential employers, who cannot directly observe the candidate's abilities.

4. Screening

Screening is an effort by the less informed party to induce the more informed party to reveal information.

Example:

Job Recruitment: Employers might require candidates to go through multiple rounds of interviews, tests, and background checks to gather more information about their skills and suitability for the job.

5. Hidden Actions and Hidden Information

These occur when one party cannot observe the actions or obtain the information of another party, leading to inefficient outcomes.

Example:

Automobile Market: Sellers of used cars typically have more information about the condition of the vehicle than buyers. This can lead to a market for "lemons," where buyers are only willing to pay an average price, knowing there is a risk of getting a substandard car. This discourages sellers with good-quality cars from entering the market, reducing overall market quality.

Impact of Informational Asymmetries

Informational asymmetries can lead to several market issues:

Market Failure: High levels of adverse selection or moral hazard can cause markets to fail, as seen in the health insurance and subprime mortgage markets.

Reduced Market Efficiency: When buyers or sellers cannot accurately assess the value of a product, transactions that would be beneficial to both parties may not occur.

Increased Costs: Firms might spend significant resources on screening or signaling to mitigate informational asymmetries.

Mitigating Informational Asymmetries

Several mechanisms can help reduce informational asymmetries:

Regulation: Governments can require disclosure of information, such as the nutritional content on food packaging or financial disclosures by publicly traded companies.

Reputation Systems: Online marketplaces like eBay and Amazon use reputation systems to help buyers and sellers gauge trustworthiness.

Third-party Verification: Independent audits and certifications can provide assurance about the quality and safety of products and services.

Understanding and addressing informational asymmetries is crucial for improving market efficiency and ensuring fair transactions.

B) Why might a labor market unravel? Discuss in brief.

Ans: A labor market might unravel due to several factors that disrupt the balance between supply and demand for labor, leading to inefficiencies and imbalances. Here are some key reasons why this can happen:

1. Informational Asymmetry

In the labor market, informational asymmetry occurs when one party (either the employer or the employee) has more or better information than the other. For example, employers may not fully know the capabilities or work ethic of potential hires, while employees might not know the true working conditions or job security.

Impact: This can lead to adverse selection, where high-quality candidates are not hired because employers cannot distinguish them from less qualified candidates, and moral hazard, where employees might shirk responsibilities if their performance is not properly monitored.

2. Mismatched Expectations

When there is a significant gap between what employers offer and what employees expect, the market can unravel. This can happen due to differences in expectations about wages, job responsibilities, career growth, and work-life balance.

Impact: Employees may leave for better opportunities, or potential candidates may not accept job offers, leading to high turnover rates and unfilled positions.

3. Labor Market Rigidity

Rigidities such as long-term contracts, minimum wage laws, and strong labor unions can prevent the labor market from adjusting to changes in supply and demand. While these mechanisms can protect workers, they can also lead to inefficiencies.

Impact: Employers may be unable to reduce labor costs during economic downturns, potentially leading to layoffs or closures. Similarly, employees may find it difficult to move to better job opportunities due to long-term contracts.

4. Skill Mismatch

When the skills of the labor force do not match the requirements of available jobs, the market can unravel. This can be due to rapid technological changes, shifts in industry demand, or inadequate education and training systems.

Impact: Employers may struggle to find suitable candidates, while job seekers may find themselves unemployed or underemployed despite available positions.

5. External Shocks

External economic shocks, such as recessions, technological advancements, or globalization, can disrupt the labor market. These shocks can lead to sudden changes in demand for certain types of labor.

Impact: Industries may contract or expand rapidly, leading to job losses in some sectors and shortages in others. Workers may need to retrain or relocate, causing temporary unemployment and market inefficiencies.

6. Search and Matching Problems

The process of matching job seekers with job vacancies can be complex and time-consuming. Imperfect information and the costs associated with job searching and recruitment can lead to delays and mismatches.

Impact: Even if there are many job openings and job seekers, they may not find each other efficiently, leading to prolonged unemployment and unfilled vacancies.

7. Institutional Failures

Failures in institutions that support the labor market, such as unemployment benefits systems, labor regulations, and education systems, can cause the market to unravel. Ineffective policies or poorly managed institutions can exacerbate unemployment and hinder job creation.

Impact: Workers may not receive adequate support during transitions, and employers may face obstacles in hiring and training workers, leading to a less dynamic labor market.

Example: The Great Recession

During the Great Recession (2007-2009), the labor market in many countries unraveled due to a combination of these factors. High levels of uncertainty, skill mismatches, and economic shocks led to massive layoffs and prolonged unemployment. Many workers had to accept lower-paying jobs or leave the labor force altogether, while employers struggled to find qualified candidates as the economy began to recover.

In summary, a labor market can unravel due to informational asymmetries, mismatched expectations, rigidities, skill mismatches, external shocks, search and matching problems, and institutional failures. Addressing these issues requires coordinated efforts from policymakers, employers, and workers to ensure a more balanced and efficient labor market.

C) In Bangladesh, bus drivers have at times been paid based on the number of riders they pick up. Discuss the likely differences in driver behavior under this plan compared to a plan in which drivers are paid by the day.

Ans: When bus drivers are paid based on the number of riders they pick up, their behavior is likely to differ significantly compared to when they are paid a fixed daily wage. Here are the key differences:

Incentives and Behavior

1. Speed and Aggressiveness:

Per Rider Payment: Drivers are incentivized to pick up as many passengers as possible, leading them to drive aggressively, make frequent stops, and compete with other buses for passengers. This can result in reckless driving, increased accidents, and traffic congestion.

Daily Wage Payment: Drivers have no financial incentive to pick up more passengers beyond what is necessary. They are more likely to drive at a steady pace, adhere to traffic rules, and avoid unnecessary competition with other buses.

2. Route Efficiency:

Per Rider Payment: Drivers may take longer, less efficient routes that maximize the number of stops and potential passengers, even if it inconveniences the current riders. This can lead to increased travel times for passengers.

Daily Wage Payment: Drivers are more likely to stick to designated routes and schedules, prioritizing efficiency and timeliness over maximizing passenger numbers.

3. Service Quality:

Per Rider Payment: The focus on picking up more passengers can compromise service quality. Drivers may overload buses, neglect maintenance, and prioritize short-term gains over passenger comfort and safety.

Daily Wage Payment: Drivers may be more consistent in providing quality service, maintaining buses properly, and ensuring passenger comfort and safety, as their income is not directly tied to the number of passengers.

4. Passenger Experience:

Per Rider Payment: Passengers may experience frequent stops, overcrowded buses, and erratic driving. This can lead to a less predictable and less comfortable travel experience.

Daily Wage Payment: Passengers are likely to experience a more predictable and stable service, with fewer stops, less overcrowding, and more consistent driving behavior.

Potential Outcomes

1. Safety Concerns:

Per Rider Payment: Increased risk of traffic accidents due to aggressive driving and competition for passengers.

Daily Wage Payment: Potentially safer driving practices, reducing the likelihood of accidents.

2. Traffic Flow:

Per Rider Payment: Increased congestion and disruptions in traffic flow due to frequent stopping and aggressive driving.

Daily Wage Payment: Smoother traffic flow and less congestion as drivers follow more consistent routes and schedules.

3. Economic Efficiency:

Per Rider Payment: Short-term economic gains for drivers but potential long-term inefficiencies due to safety and congestion issues.

Daily Wage Payment: More stable and potentially more efficient system, with long-term benefits for both drivers and passengers.

4. Driver Income Stability:

Per Rider Payment: Income for drivers may be highly variable, depending on passenger numbers, which can lead to financial instability.

Daily Wage Payment: Provides drivers with a stable and predictable income, reducing financial stress.

Example from Bangladesh

In Bangladesh, bus drivers often face challenging working conditions, and their compensation structures can significantly impact their behavior. When drivers are paid based on the number of riders, they may prioritize maximizing their income over safety and service quality. This can exacerbate existing issues such as traffic congestion, road safety concerns, and poor public transportation experiences.

Conversely, a shift to a daily wage payment structure could help improve these issues by aligning drivers' incentives with safer driving practices and more reliable service. However, this

transition would require careful planning and enforcement to ensure that service standards are maintained and drivers receive fair compensation.

In conclusion, the method of compensation has a profound impact on driver behavior. Paying drivers based on the number of riders can lead to aggressive and unsafe driving, while paying drivers a fixed daily wage can promote safer, more consistent, and higher-quality service.